

Practice set on combinatorics

Disclaimer: These questions have been taken from multiple sources and all the details have not been verified due to time constraints. So, there might be typographical and other errors. Also, the difficulty level of these questions have nothing to do with the difficulty level of the exam.

1. Independent Cartesian choices: You design a combo by choosing 1 of 7 appetizers, 1 of 9 main courses, and 1 of 5 desserts. How many combos?
2. Sequential choices without replacement: From 12 books, pick an ordered reading list of 3 distinct books. How many ordered triples?
3. Dependent choices with forbidden adjacency: Build a 5-character code from letters A–F so that no two adjacent characters are equal. How many codes?
4. Conditional second-stage choices: Choose a chair (10 options). If you choose chair #1, choose a cushion from 4 colors; otherwise choose from 7 colors. How many chair–cushion pairs?
5. Multi-role assignment without reuse: Form a 4-person committee of Chair, Secretary, Treasurer, and Member from 10 people. Each person may occupy only one role. How many assignments?
6. Position-dependent alphabets: Form a 6-character string where positions 1–3 are vowels (5 choices each) and positions 4–6 are digits (10 choices each). How many strings?
7. Color-matching constraint: Pick a pair (shirt, tie). There are 8 shirt colors. Each shirt color has exactly 2 ties whose color name begins with the same letter. How many valid pairs?
8. Layered choices with varying branch sizes: There are 4 car models. Their trim options are 3, 4, 2, and 5 respectively. Choose a model and then a trim. How many possibilities?
9. Circular layout with fixed starting reference: Six plants are placed around a circle with a fixed marked starting plant. Assign to each plant one of 26 tag letters (repetition allowed). How many assignments?
10. Independent choices for multiple identical containers: You prepare 4 identical gift boxes. For each box independently choose one of 10 toy types (repetition allowed). How many distributions?
11. Choice-tree with different branch sizes: First choose a country (5 choices). If you choose A, then choose a city from 3 options; if B, from 7 options; otherwise from 4 options. How many ordered pairs?

12. Exclusion rule tied to a previous choice: Choose a 3-digit PIN. The first digit cannot be 0. If the first digit is 7, then the second digit cannot be 3 (otherwise it has 10 choices). How many PINs?
14. A 4-character code is formed as follows: the first character is a digit (0–9). If the first character is even, the second character is an uppercase letter (26 choices); otherwise the second character is a lowercase letter (26 choices). The last two characters may be any of the 52 letters. How many codes can be formed?
15. A robot chooses a movement pattern: first choose a starting direction from $\{N, E, S, W\}$. If N or S is chosen, choose one of 5 speeds; if E or W , choose one of 3 speeds. Then choose a sequence of 4 moves, each move being one of 6 actions. How many distinct movement patterns?
16. A password is created in three phases. Phase 1: pick a symbol from a set of 12. Phase 2: if the symbol is from the subset $\{\#, @, *\}$, pick a 3-letter string from uppercase letters; otherwise pick a 2-letter string from lowercase letters. Phase 3: append a number. If Phase 2 used 3 letters, pick any 2-digit number; if Phase 2 used 2 letters, pick any 3-digit number. How many passwords?
17. A manufacturing line builds devices in stages. Stage 1: choose a chip type A (10 variants) or B (6 variants). Stage 2: if chip type A is chosen, choose one of 4 compatible casings; if B , choose one of 7 casings. Stage 3: choose a battery. Casings 1–2 of type A allow 5 batteries each; casings 3–4 allow 3 batteries each. All B -casings allow 4 batteries. Stage 4: choose a software version from 12. How many distinct devices can be manufactured?
18. A student fills a 5-question adaptive test. Q1: choose one of 4 modules. If module $M1$ is chosen, Q2 has 3 options; if $M2$ or $M3$, Q2 has 5 options; if $M4$, Q2 has 2 options. For Q3–Q5: - If Q2's option number is even, each question has 4 choices. - If Q2's option number is odd, each has 6 choices. Count the total number of distinct answer sequences.
19. Consider a 7-step experiment. Step 1: choose a parameter p from $\{1, 2, 3, 4, 5\}$. Steps 2–4: if p is prime, each of these steps has 8 outcomes; otherwise each has 5 outcomes. Steps 5–7: if the product of outcomes in steps 2–4 is even, each step has 6 outcomes; otherwise each step has 9 outcomes. How many total outcomes does the experiment have?
20. A secure access protocol works as follows: Step 1: choose an authentication mode A , B , or C . Step 2: - If A , choose a 4-bit key (16 options). - If B , choose a pair of distinct keys from a pool of 10. - If C , choose an ordered triple (k_1, k_2, k_3) of keys from a pool of 6 keys, with repetition allowed. Step 3: the system then assigns a challenge depending on the key(s) chosen. - If the key set has size 1, there are 8 challenges. - If the key set has size 2, there are 20 challenges. - If the key set has size 3, there are 30 challenges. Count the number of total authentication processes.
21. A hybrid machine-learning pipeline is chosen as follows: Select a model type: Decision Tree (3 variants), Neural Network (5 variants), or Transformer (4 variants). For Decision Trees: choose depth $d \in \{3, 4, 5, 6\}$, then choose $2d$ hyperparameters, each from 7 values. For Neural Networks: choose number of layers $\ell \in \{3, 4, 5\}$. If ℓ is even, choose 3ℓ hyperparameters from 10 values each; otherwise choose 2ℓ hyperparameters from 12 values each. For Transformers:

choose number of heads $h \in \{4, 8\}$ and embedding dimension $e \in \{64, 128, 256\}$. Then choose $h + e/64$ hyperparameters from 9 values each. How many total pipelines?

22. A 10-step algorithm behaves as follows. Step 1: choose a mode X, Y , or Z . Step 2: - If X , choose a number from 1–4. - If Y , choose a number from 1–6. - If Z , choose a number from 1–9. Steps 3–6: - If Step 2's number is in $\{1, 2\}$, each step has 3 actions. - If it is in $\{3, 4, 5, 6\}$, each has 5 actions. - Otherwise each has 7 actions. Steps 7–10: - If the total number of actions chosen in Steps 3–6 is congruent to 0 mod 3, each step has 4 options; - otherwise each has 6 options. Compute the total number of possible executions.
23. A hierarchical network is designed in four layers. Layer 1: choose one of 5 protocols. Layer 2: - If Protocol 1 or 2 is chosen, select a pair of distinct nodes from 12. - If Protocol 3 is chosen, select an ordered pair from 12. - If Protocol 4 or 5 is chosen, select an ordered triple from 8 with repetition allowed. Layer 3: - If the number of selected nodes is 2, choose any subset of size 3 from 10 resources. - If the number of selected nodes is 3, choose any subset of size 4 from 10 resources. Layer 4: choose a routing pattern from 15 options. How many distinct network configurations exist?
25. How many binary strings of length n contain an even number of 1s?
26. In how many ways can n people be seated around a round table?
27. How many partitions of an integer n consist only of odd parts?
28. How many ways to choose k elements from $[n]$ and then choose a superset of that chosen set?
29. How many ways can you choose a committee from n people and then choose a chairperson from that committee?
30. How many proper colorings of a path graph P_n using q colors are possible?
31. How many subsets of $[n]$ have largest element exactly m ?
32. How many binary strings of length n start and end with different bits?
33. How many ternary strings of length n have exactly k zeros?
34. How many ways to distribute k identical balls into n distinct boxes?
35. How many sequences of length r from $[n]$ are weakly increasing?
36. How many subsets of $[n]$ have at least one even number?
37. How many functions $f : [n] \rightarrow [k]$ have image size exactly r ?
38. How many permutations of $[n]$ have no fixed points except possibly at position 1?
39. How many words of length n over an alphabet of size q have all adjacent characters distinct?
40. How many subsets of $[n]$ have exactly k elements and contain the element m ?
41. How many ways to choose a nonempty subset of $[n]$ and then pick one distinguished element from within it?

45. How many distinct necklaces with 6 beads can be made from 3 colors if rotations are considered identical?
46. How many distinct 5-letter words can be formed from the letters A, B, C, D, E if words that differ only by a cyclic shift are considered the same?
47. How many ways can 8 people sit around a round table if rotations are considered identical?
48. A bracelet has 7 beads and each bead can be red, blue, or green. How many distinct bracelets exist if flipping the bracelet counts as the same?
49. How many distinct arrangements of the letters in the word "BANANA" are there if swapping identical letters is not counted as creating a new arrangement?
50. How many distinct ways are there to seat 12 people at a round table if seating arrangements that are rotations or reflections of each other are considered identical?
51. How many distinct circular arrangements of 10 colored flags are there if arrangements that are rotations of each other are considered identical?
52. A password consists of 6 characters chosen from $\{A, B, C, D\}$. Two passwords are considered equivalent if one can be obtained from the other by a rotation of characters. How many distinct passwords are there?
53. How many different colorings of the faces of a cube are possible using 3 colors if rotations of the cube do not produce a new coloring?
54. How many distinct circular necklaces with 8 beads can be made using exactly 2 red beads and 6 blue beads if rotations are considered identical?
55. How many distinct ways can 5 identical coins be arranged on a circular plate if arrangements that differ by rotation are considered the same?
56. How many distinct arrangements are there of the letters of the word "LEVEL" if arrangements that differ only by reflection are considered identical?
57. How many distinct circular arrangements are there of 9 people in a circle if rotations are considered identical, but reflections are counted as different?
58. How many ways are there to arrange 7 different books on a circular shelf if rotations are considered identical?
59. How many distinct bracelets with 5 beads can be made from 4 colors if rotations but not reflections are considered identical?
60. How many subsets does a set with 7 elements have?
61. How many subsets of size 3 can be formed from a set of 8 elements?
62. How many subsets of $\{1, 2, 3, 4, 5\}$ contain the element 1?
63. A set has 6 elements. How many subsets contain at least one element?

65. How many subsets of $\{1, 2, 3, 4\}$ contain exactly 2 elements?
66. How many ways can a set of 10 elements be split into 2 disjoint subsets of sizes 3 and 7?
67. How many ways can a set of 8 elements be partitioned into 3 nonempty subsets?
68. How many subsets of $\{1, 2, \dots, 9\}$ contain either element 2 or element 5 (or both)?
69. How many subsets of size 4 can be formed from a set of 12 elements such that two specific elements are not together in the subset?
70. A set of 7 elements is split into two subsets: one of size 3 and the other containing the remaining elements. How many ways can this be done?
71. How many subsets of a 6-element set have an even number of elements?
72. A set has 5 elements. How many ordered pairs of disjoint subsets can be formed?
73. A set of 12 elements is split into 3 subsets of sizes 4, 4, and 4. How many ways can this be done?
74. How many subsets of $\{1, 2, \dots, 10\}$ have more than 3 elements but fewer than 7?
75. How many ways can a 10-element set be split into 4 subsets of sizes 2, 3, 2, and 3?
76. How many subsets of $\{1, 2, \dots, 8\}$ contain exactly 3 elements from the first 5 elements?
77. A set of 9 elements is split into 3 subsets of sizes 2, 3, and 4. How many ways can this be done if the order of subsets matters?
78. How many subsets of $\{1, 2, \dots, 10\}$ contain an equal number of odd and even numbers?
79. How many ways can a 12-element set be split into 3 subsets of sizes 3, 4, and 5, with the largest element always in the largest subset?
80. How many subsets of a 10-element set contain at least 1 element but no more than 5 elements?
81. A set of 15 elements is split into 5 subsets of size 3 each. How many ways can this be done if the order of subsets matters?
82. How many subsets of $\{1, 2, \dots, 12\}$ have exactly 4 elements, with no two consecutive numbers?
83. A set of 10 elements is split into 2 subsets such that one subset has at least 3 elements and the other has at most 7 elements. How many ways can this be done?
84. How many subsets of $\{1, 2, \dots, 10\}$ contain exactly 2 odd numbers and 2 even numbers?
85. How many ways can a set of 9 elements be split into 3 non-empty subsets such that the smallest element is always in the first subset?
86. How many subsets of $\{1, 2, \dots, 12\}$ contain at least 3 elements and exclude elements 1 and 2?
87. A set of 14 elements is split into 4 subsets of sizes 2, 3, 4, and 5. How many ways can this be done?

88. How many subsets of a 12-element set contain exactly 5 elements, none of which are consecutive integers?
89. How many ways can a 10-element set be split into 3 subsets of sizes 2, 3, and 5 such that the largest element is in the largest subset and the smallest element is in the smallest subset?
90. How many subsets of $\{1, 2, \dots, 15\}$ contain exactly 7 elements, with at least 3 elements being multiples of 3?
92. How many 5-digit numbers have distinct digits?
93. In a class of 50 students, show that at least 5 students have last names starting with the same letter.
94. A set has 6 elements. How many ordered pairs of disjoint subsets (A, B) can be formed?
95. A password consists of 2 letters followed by 3 digits. How many passwords are possible?
96. How many integers between 1 and 200 are divisible by 2, 3, or 5?
97. How many sequences of length 5 can be formed using the digits 1–5 if repetition is allowed but no sequence is strictly increasing?
98. How many socks must you pick from a drawer containing 10 black, 12 white, and 15 blue socks to guarantee at least 5 of the same color?
99. How many 4-digit numbers can be formed from $\{0, 1, 2, 3, 4, 5\}$ if digits cannot repeat and the first digit is non-zero?
100. How many numbers between 1 and 500 are divisible by 3 or 4 but not by 5?
101. A student has 3 shirts, 2 pants, and 4 pairs of shoes. How many different outfits can be made?
102. How many subsets of a 10-element set contain at least one element from a specified subset of 4 elements?
103. A set of 10 elements is split into two subsets of sizes 4 and 6. How many ways can this be done?
104. In a group of 100 people, prove that at least 2 people have the same number of friends within the group.
105. How many subsets of size 3 or 4 can be formed from a set of 8 elements?
106. How many integers between 1 and 1000 are not divisible by 2, 3, or 5?
107. How many 7-digit numbers can be formed using digits 1–5 if each digit appears at least once?
108. From a group of 6 men and 5 women, in how many ways can a team of 4 be chosen such that at least 2 are women?
109. How many strings of length 6 can be formed using the letters A, B, C, D if A must appear at least once?

110. How many integers between 1 and 1000 are divisible by exactly one of 2, 3, or 5?
111. In a class of 30 students, how many ways can you choose a subset such that no two students have consecutive roll numbers?
112. How many ways can a set of 10 elements be split into two subsets such that one subset has at least 3 elements and the other has at most 7 elements?
113. How many sequences of length 5 can be formed using digits 1–5 if repetition is allowed and each digit appears at least once?
114. Show that in any group of 13 people, at least two were born in the same month.
115. In how many permutations of $\{1, 2, 3, 4, 5, 6\}$ does at least one element stay in its original position?
116. A committee of 3 is selected from 5 men and 4 women. How many ways can the committee have at least 1 woman?
118. Show that $\sum_{k=0}^n \binom{n}{k} \binom{k}{r} \binom{n-k}{s} = \binom{n}{r+s} \binom{r+s}{r} 2^{n-r-s}$.
119. Prove that $\sum_{k=0}^n k \binom{n}{k} = n \cdot 2^{n-1}$.
120. Prove Pascal's Identity: $\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$.
121. Show that $\sum_{k=0}^n k \binom{n}{k}^2 = n \binom{2n-1}{n-1}$.
122. Prove that $\sum_{k=0}^n \binom{n}{k}^3 = \sum_{k=0}^n \binom{n}{k} \binom{n}{n-k} \binom{2n-k}{n}$.
123. Prove that $\binom{n}{k} = \binom{n}{n-k}$.
124. Prove that $\sum_{k=0}^n k^3 \binom{n}{k} = \frac{n^2(n+1)2^{n-2}}{2}$.
125. Show that $\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}$.
126. Show that $\sum_{i=0}^k (-1)^i \binom{n}{i} = (-1)^k \binom{n-1}{k}$.
127. Prove that $\sum_{k=0}^n k \binom{n}{k} \binom{n-k}{r} = (r+1) \binom{2n-r-1}{n-1}$.
128. Show that $\sum_{k=0}^n \binom{n}{k} = 2^n$.
129. Prove that $\sum_{k=0}^n \binom{n}{k} k^2 = n(n+1)2^{n-2}$.
130. Show that $\sum_{k=0}^n k \binom{n}{k} \binom{n-1}{k-1} = \sum_{k=0}^n k \binom{n}{k}$.
131. Prove that $\sum_{k=0}^n (-1)^k \binom{n}{k}^2 = 0$ for $n \geq 1$.
132. Prove that $\sum_{k=1}^n \binom{n}{k} = 2^n - 1$.
133. Prove that $\sum_{k=0}^n \binom{n}{k} \binom{n-k}{r} = \binom{n}{r} 2^{n-r}$.

134. Show that $\sum_{k=0}^n (-1)^k \binom{n}{k} 2^{n-k} = 1$.
135. Prove that $\sum_{k=0}^n \binom{n}{k} \binom{m}{r-k} = \binom{n+m}{r}$ (Vandermonde's Identity).
136. Show that $\sum_{k=0}^n k^2 \binom{n}{k} = n(n+1)2^{n-2}$.
137. Prove that $\sum_{k=0}^n (-1)^k \binom{n}{k} (n-k)^m = 0$ for $0 \leq m < n$.
138. Show that $\sum_{k=0}^n \binom{n}{k} k^3 = n^2 2^{n-2} (n+1)/2$.
139. Prove that $\binom{n}{k} \binom{k}{r} = \binom{n}{r} \binom{n-r}{k-r}$.
140. Show that $\sum_{k=0}^n \binom{n}{k} \binom{2k}{k} = \binom{2n}{n}$.
141. Show that $\sum_{k=0}^n \binom{n}{k} \binom{n+k}{k} = \binom{2n}{n}$.
142. Prove that $\sum_{k=0}^n k \binom{n}{k}^3 = n^2 \binom{2n-1}{n-1}$.